

SyncDrive

Signal-Aware Navigation and Speed Optimization System
Project Proposal for UCS503P
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1 Higher-order goal

... secondary framing

SyncDrive aims to reduce avoidable time and fuel loss in everyday urban commuting by making drivers aware of how their speed interacts with upcoming traffic signals. The engineering objective is to translate *signal-aware speed guidance* into a measurable, deployable software solution usable through a navigation application.

2 Time-to-value

... primary heuristic

- Deliver meaningful increments quickly by prototyping the core route-and-speed-recommendation workflow on a single simulated corridor within a short iteration window.
- Prioritize fast feedback over perfection using weekly iterations and early CI-backed automation.
- Success in iteration one is a working speed recommendation for a route with simulated signals, improved thereafter based on testing.

3 Problem Statement

Navigation apps such as Google Maps select routes using distance, travel time, and traffic, but do not account for how a driver's chosen speed interacts with signal timing.

- **Signal-blind routing:** a short, lightly congested route can still force repeated stops if driven at the speed limit.
- **Avoidable stop-and-go driving:** repeated braking and waiting increases journey time even in light traffic.
- **No dynamic speed guidance:** apps advise *which road* to take, not *how fast* to drive on it.
- **No driver-level emergency coordination:** nearby drivers get no software guidance to help clear a path for an ambulance.

Why this matters now: in corridors with closely spaced signals, reducing unnecessary stopping can measurably cut journey time without changing road infrastructure.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **web/mobile-based** navigation system with:

- **Driver interface:** source/destination entry; route view with upcoming signals.
- **Signal timing store:** simulated/map-sourced signal location, cycle duration, expected green/red intervals.
- **Speed optimization engine:** computes a dynamically updated, legal recommended speed to increase the likelihood of green-phase arrivals.

- **Recommendation display:** e.g., “Maintain 42 km/h for the next 800 m,” always capped at the speed limit.
- **Emergency coordination module:** adjusts nearby drivers’ recommendations when a registered emergency vehicle approaches.

4.2 Core workflow

1. Driver enters source/destination; a route is selected using distance, time, and traffic.
2. System identifies signals along the route and retrieves simulated/known timing.
3. Engine estimates arrival time per candidate speed and selects the legal speed minimizing expected waiting time across the route.
4. Recommendation recalculates as position, traffic, or emergency conditions change.

4.3 Operational constraints

- Recommended speed never exceeds the legal limit – the system chooses *which* legal speed, not whether to speed.
- No guarantee of a green signal at every intersection; guidance is expressed as increased probability and reduced expected waiting time.
- Real-time municipal signal-controller access is not assumed; simulated/public data is used and this limitation is stated explicitly.

5 Solution Approach

... *Engineering focus*

This is an engineering project, so we focus on a realistic, implementable design rather than novel traffic-prediction research.

5.1 Speed optimization logic

... *non-research*

For each upcoming signal:

- Determine distance from the current position to the signal.
- Retrieve the signal’s current/predicted phase and cycle timing.
- Estimate arrival time for candidate legal speeds ($arrivaltime = distance/speed$).
- Estimate likely phase (green/red) and expected waiting time per candidate.
- Select the legal speed minimizing a weighted cost over travel time, waiting time, and stop count, considering downstream signals.
- Recompute whenever position, traffic, or signal data changes materially.

This is a prediction heuristic, not a guarantee of signal outcomes or a claim of novel research.

5.2 System architecture

... *web-based solution*

- **Frontend:** map, route, signal markers, recommended speed, ETA, emergency notifications.
- **Backend API:** route requests, signal-timing queries, optimization calculations, emergency events.
- **Route/traffic services:** a mapping/routing API (e.g., OpenStreetMap-based, Mapbox) for road network, route, distance, coordinates; signal timing is simulated separately since such APIs rarely expose real-time signal-phase data.
- **Database:** routes, simulated signal timing, requests, emergency events.

5.3 CI/CD and fast delivery enabler

- Build and test automatically on every change, including the optimization logic.
- Produce repeatable deployment artifacts to staging.
- Enable safe, frequent releases of incremental features (multi-signal optimization, emergency coordination).

6 Evaluation Criterion

... *measurable and attributable*

Success is measured by comparing the signal-aware strategy against a conventional constant-speed-limit strategy on identical simulated routes.

6.1 Primary evaluation metric

Total journey time: time to traverse a test route source-to-destination. Objective: reduce total journey time relative to the conventional strategy, measured experimentally. Attribution: simulated start/end timestamps under identical test conditions.

6.2 Secondary evaluation metrics

- Average signal waiting time and number of complete stops per route.
- Percentage of signals crossed during green phases.
- Average speed and travel distance across strategies.
- Emergency-vehicle travel-time improvement in the simulated coordination scenario.

No numerical improvement is assumed in advance; the prototype measures and compares these metrics experimentally.

6.3 Pilot validation plan

... high-level

- Define 3–5 test routes of varying length and signal density using simulated timing data.
- Run each route under both strategies and log all metrics.
- Gather informal driver feedback (5–10 participants) on clarity and usefulness.

7 Scalability

... with foresight

1. **Scalable (theoretically):** decoupled frontend/backend/optimization engine; indexed queries for signals/routes; stateless API design; caching of frequent route/signal data.
2. **Operationally deployable (practically):** managed database; containerized/platform-hosted deployment; CI/CD pipeline; modular services so signal, route, and emergency data scale independently.

8 Engine availability heuristic

Mature tooling is available for this project: a web/mobile frontend framework; a backend REST framework; a mapping/routing API (road network, route, distance, coordinates – not assumed to expose real-time signal phase); geolocation services; a managed database; a simulated signal-timing dataset built for the prototype; a test framework; and a CI/CD runner. We rely on these mature components to focus on optimization-logic correctness and workflow, not on inventing infrastructure or assuming access to real municipal signal controllers.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- Source/destination input and basic map interface; route generation via a routing API.
- Simulated traffic-signal dataset for a test corridor.
- Basic arrival-time calculation and speed recommendation for a single upcoming signal, capped at the speed limit.
- Basic backend API with unit tests for the optimization logic; CI build + tests + linting.

9.2 Subsequent deliverables

- Multi-signal optimization considering downstream signals.
- Dynamic recalculation as the vehicle moves or conditions change.
- Emergency-vehicle coordination module (simulated ambulance location, adjusted nearby recommendations).
- Dashboard comparing signal-aware vs. conventional strategy; performance/metric logging; scalability improvements (caching, indexing).

10 Risks and mitigations

- **Signal data availability:** real-time phase data is rarely public. *Mitigation:* simulated/controlled data, limitation stated clearly.

- **Prediction uncertainty:** actual timing may differ from predicted. *Mitigation:* express expected/estimated waiting time, not a guarantee; recalculate continuously.
- **Mapping API limitations:** usage limits or missing traffic data. *Mitigation:* fall back to open/simulated signal data.
- **Safety:** must never encourage exceeding the speed limit. *Mitigation:* engine constrains all speeds to the configured legal limit.
- **Emergency-coordination realism:** real infrastructure/control is inaccessible. *Mitigation:* simulated emergency-vehicle environment, framed as future-integrable rather than working today.
- **Evaluation limitations:** a small prototype cannot reproduce city-scale traffic. *Mitigation:* controlled simulations with clearly defined test scenarios.

11 Summary

SyncDrive proposes a deployable navigation system that recommends a legal driving speed to reduce unnecessary signal stops, alongside conventional route selection. It emphasizes time-to-value through rapid iterations, CI/CD-backed releases, and measurable evaluation (primarily total journey time, plus waiting-time and stop-count metrics), while remaining realistic about data limitations and explicitly avoiding claims of guaranteed green signals, guaranteed ambulance priority, or control of real traffic infrastructure.